

Practical -12

```
himasree@LAPTOP-41COT5AA:~/OSLab$ mkdir -p ~/os_sync_project
cd ~/os_sync_project
himasree@LAPTOP-41COT5AA:~/os_sync_project$ pwd
/home/himasree/os_sync_project
himasree@LAPTOP-41COT5AA:~/os_sync_project$ touch main.c producer_consumer.c producer_consumer.h deadlock.c deadlock.h Makefile
himasree@LAPTOP-41COT5AA:~/os_sync_project$ ls
Makefile deadlock.c deadlock.h main.c producer_consumer.c producer_consumer.h
himasree@LAPTOP-41COT5AA:~/os_sync_project$ nano producer_consumer.h
himasree@LAPTOP-41COT5AA:~/os_sync_project$ nano producer_consumer.c
himasree@LAPTOP-41COT5AA:~/os_sync_project$ nano main.c
himasree@LAPTOP-41COT5AA:~/os_sync_project$ gcc main.c producer_consumer.c -o os_sync -pthread
himasree@LAPTOP-41COT5AA:~/os_sync_project$ ls
Makefile deadlock.c deadlock.h main.c os_sync producer_consumer.c producer_consumer.h
himasree@LAPTOP-41COT5AA:~/os_sync_project$ ./os_sync

=====
OS SYNCHRONIZATION PROJECT
=====
1. Producer-Consumer
2. Exit
Enter choice: |
```

```
=====
OS SYNCHRONIZATION PROJECT
=====
1. Producer-Consumer
2. Exit
Enter choice: 1
Enter buffer size: 5

=====
PRODUCER-CONSUMER DEMONSTRATION
=====
Buffer size: 5

[Producer] Produced item 1 at buffer position 0
[Consumer] Consumed item 1 from buffer position 0
[Producer] Produced item 2 at buffer position 1
[Consumer] Consumed item 2 from buffer position 1
[Producer] Produced item 3 at buffer position 2
[Producer] Produced item 4 at buffer position 3
[Consumer] Consumed item 3 from buffer position 2
[Producer] Produced item 5 at buffer position 4
[Consumer] Consumed item 4 from buffer position 3
[Producer] Produced item 6 at buffer position 0
[Consumer] Consumed item 5 from buffer position 4
[Producer] Produced item 7 at buffer position 1
[Producer] Produced item 8 at buffer position 2
[Consumer] Consumed item 6 from buffer position 0
```

```
Synchronization check: PASSED
All 20 items were produced and consumed.
```

```
=====
OS SYNCHRONIZATION PROJECT
=====
1. Producer-Consumer
2. Exit
Enter choice: |
```

```
himasree@LAPTOP-41COT5AA:~/os_sync_project$ gcc main.c producer_consumer.c deadlock.c -o os_sync -pthread
himasree@LAPTOP-41COT5AA:~/os_sync_project$ ./os_sync

=====
OS SYNCHRONIZATION PROJECT
=====
1. Producer-Consumer
2. Demonstrate Deadlock
3. Prevent Deadlock
4. Exit
Enter choice: 2

=====
DEADLOCK DEMONSTRATION
=====

Resources:
Resource A
Resource B

Thread 1 will acquire A and then request B.
Thread 2 will acquire B and then request A.

Starting threads...

[Thread 1] Started
[Thread 1] Trying to acquire Resource A...
[Thread 1] Resource A acquired.
```

```
=====
DEADLOCK OCCURRED
=====
```

```
Thread 1:
  Holds Resource A
  Waiting for Resource B
```

```
Thread 2:
  Holds Resource B
  Waiting for Resource A
```

Four necessary conditions:

1. Mutual Exclusion : YES
2. Hold and Wait : YES
3. No Preemption : YES
4. Circular Wait : YES

The threads are intentionally left waiting.
Returning to the main program without joining them.

```

himasree@LAPTOP-41COT5AA:~/os_sync_project$ nano main.c
himasree@LAPTOP-41COT5AA:~/os_sync_project$ gcc main.c producer_consumer.c deadlock.c -o os_sync -pthread
himasree@LAPTOP-41COT5AA:~/os_sync_project$ ./os_sync

=====
OS SYNCHRONIZATION PROJECT
=====
1. Producer-Consumer Demonstration
2. Throughput Evaluation
3. Demonstrate Deadlock
4. Prevent Deadlock
5. Exit
-----
Enter choice: 3

=====
DEADLOCK DEMONSTRATION
=====

Resources:
Resource A
Resource B

Thread 1 will acquire A and then request B.
Thread 2 will acquire B and then request A.

Starting threads...

```

```

=====
DEADLOCK OCCURRED
=====

Thread 1:
  Holds Resource A
  Waiting for Resource B

Thread 2:
  Holds Resource B
  Waiting for Resource A

Four necessary conditions:
1. Mutual Exclusion    : YES
2. Hold and Wait      : YES
3. No Preemption      : YES
4. Circular Wait      : YES

The threads are intentionally left waiting.
Returning to the main program without joining them.
|

```